UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

© Hak Cipta Universiti Teknologi MARA

PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following Figure 1.
Inventor 'Y' was France's most
celebrated
mathematician
and
physicist. In 1642, inventor 'Y'
invented
and
build
the
first
working calculating machine as a
U^jj,
means of helping his father to
perform tedious tax accounting.
Figure 1: Inventor 'Y' and his calculating machine.
a)
Who is inventor 'Y'?
b)
Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.
c)
Which programming language is named after inventor 'Y'?
d)
State TWO (2) mathematical operationsthat can be performed byinventor Y's calculating machine.

Source: CSC429 (13).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Calculate -31 - (-12) using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

Answer: left -31 -> 1 0011111; right -12 -> 1 0001100; -31 - (-12) = -19 -> sign-magnitude 1 0010011

- *1 mark for converting both operands to sign-magnitude.*
- *1 mark for applying the sign rule for subtraction.*
- *1 mark for the correct magnitude operation.*
- *1 mark for the correct sign bit.*
- *1 mark for the final decimal and byte representation.*

QUESTION A3 (5 marks)

Topic: Parallel and multiprocessor architecture

a) How classification of parallel processor is constructed?
b) Give ONE (1) example for each parallel processor architecture.

Source: CSC429 (8).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

© Hak Cipta Universiti Teknologi MARA

Given the following Figure 2.

Instruction
Opcode
Address A
Memory
Operand

Figure 2

Label the addressing technique illustrated in Figure 2.

a)

Explain how the addressing technique in 9(a) is used.

b)

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

a) Describe THREE (3) types of interfaces to external devices.
b) Explain TWO (2) types of interrupt.

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.

a) Convert -6.75 to IEEE 754 single precision. Give the answer in hexadecimal.

Answer: -6.7510 = 0xC0D80000 (sign=1, exponent=10000001, fraction=10110000000000000000000)

- *1 mark for sign, exponent and mantissa.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the 32-bit binary grouping.*
- *1 mark for the hexadecimal result.*

b) Convert the IEEE 754 single-precision value 0x42C90000 to decimal.

Answer: 0x42C90000 = 100.510

- *1 mark for the sign bit.*
- *1 mark for the exponent field.*
- *1 mark for the fraction field.*
- *1 mark for applying the bias and significand.*
- *1 mark for the final decimal value.*

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

a) When is an I/O module installed?
b) What will happen if the I/O module is not installed?
c) Explain ONE (1) category of I/O module
d) What does it mean by 'hot connection'?
e) List THREE (3) common interfaces to external devices.

Source: CSC429 (21).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

a) Differentiate between organization and architecture.

© Hak Cipta Universiti Teknologi MARA

b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

Source: CSC429 (6).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Memory hierarchy and cache

a) Differentiate SRAM and DRAM from the perspective of the following characteristics,
i)
The way the bits are stored.
ii)
Speed.
iii)
Location it is being used.
b) Why does DRAM need refreshing?
c) Comparing DRAM and SRAM, which memory would be placed higher in the memory hierarchy? Give your justification to support your answer.

Source: CSC429 (14).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Computer architecture and CPU organisation

Computer consists of four main components as illustrated in Figure 1. Each component has its own function in ensuring that the computer system works well. Question 4 is based on Figure 1.

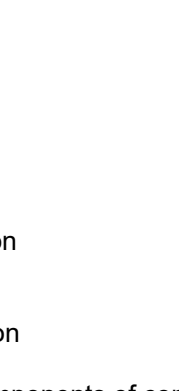


Figure 1 : Components of computer

a) Explain THREE (3) types of system interconnection.
b) State TWO (2) sub-components of central processing unit and its function.

Source: CSC429 (3).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

© Hak Cipta Universiti Teknologi MARA

PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.
y = A(A + 2B/C-D 2)
Figure 2
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.
Table 1: Machine Instruction
One-address
Three-address
Zero-address
PUSHp
LOADp
ADD (p,q,r)
POPp
STORp
SUB (p,q,r)
ADD
ADDp
MUL (p.q.r)
SUB
SUBp
DIV(p,q,r)
MUL
MULp
DIV
DIVp

Source: CSC429 (17).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

© Hak Cipta Universiti Teknologi MARA